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Computer Science

The Role of Camera-Based rPPG in Musical Biofeedback Systems for Stress Reduction and Consciousness Regulation

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Submission Date: 20. Juli 2025

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Abstract

Chronic stress in today's digital workspace often leads to a state of allostatic overload. This is a state where the body's internal warning signals become a non-conscious habit. This autopilot effect means that signals often bypass a person's conscious awareness. Traditional methods for stress regulation usually rely on passive relaxation or uncomfortable wearable sensors, which often limits how much users actually want to use them in the real world. This thesis suggests a non-contact musical biofeedback system as a more practical solution for stress and consciousness regulation.
The proposed solution uses remote Photoplethysmography (rPPG), specifically the CHROM algorithm, to track Heart Rate Variability (HRV) with a standard webcam. Compared to complex deep learning models, CHROM offers the low-latency and computational efficiency that is needed for real-time use, while staying robust against motion and different skin tones. These physiological data are then converted into an adaptive nature-based soundscape. It then acts as an external mirror of the user's internal state.
The findings suggest that this feedback helps the brain to stop overthinking and focus back on the immediate bodily experience. By intentionally interacting with the soundscape through their breathing, users can reclaim their focus and steer their autonomic nervous system back toward a healthy balance. In conclusion, camera-based musical biofeedback is a usable and scientifically grounded tool. It connects human-computer interaction with mental well-being and offers a pathway for mindfulness and regeneration in everyday life.

Contents

1 Introduction **1**
1.1 The problem: The rising need for non-invasive stress regulation 1
1.2 The concept of Biofeedback 1
1.3 The concept: Music lowering stress levels 2
1.4 Objective of this thesis 2
1.5 Structure of this Thesis 2

2 Theoretical Foundations **3**
2.1 Stress and the Autonomic Nervous System (ANS) 3
2.2 Heart Beat and Heart Rate Variability 3
2.3 Remote Photoplethysmography (rPPG) 4
2.3.1 The foundations of rPPG 4
2.3.2 Evolution: From Classic Algorithms to AI-based Detection . . 5
2.3.3 Tradtional Frameworks and Region of Interest (ROI) 5
2.3.4 Chrom approach 6
2.3.5 Modern Deep Learning Paradigms 7
2.3.6 Summary of rPPG Frameworks 8
2.4 Music and Stress Reduction 8
2.5 Consciousness and State Regulation 9
2.6 Biofeedback-Driven Regeneration 10
2.6.1 The Principle of Resonance Frequency 11
2.6.2 Autonomic Homeostasis and the Baroreflex 11
2.6.3 The Vagus Nerve and the "Braking System" 11
2.6.4 Cognitive Anchoring and Altered Consciousness 11

3 State of Art: Musical Biofeedback Systems and Empiral Evidence **12**
3.1 Case Study: The "Unwind" Musical Biofeedback System 12
3.1.1 Concept and Design Philosophy 12
3.1.2 The Multi-Layered Mapping Strategy 12
3.1.3 Experimental Results and Validation 13
3.2 Analysis of Music as a Biofeedback Signal 13
3.2.1 Study Design and Methodology 13
3.2.2 Technical Implementation: Mapping and Smoothing 14
3.2.3 Key Findings and Participant Behavior 14
3.2.4 Separating Respiration from Heart Rate 14
3.2.5 Conclusion: Why Music Matters 15
3.3 Case Study: Technology-Mediated Mindfulness 15
3.3.1 Problem Statement and Objectives 15
3.3.2 Methodology and Experimental Design 15
3.3.3 Muse headband 16
3.3.4 Analysis of Results: Psychological and Cognitive Effects 16

3.3.5 Conclusion: Technology as a Motivational Catalyst 16

4 Discussion **18**
4.1 Using the heartbeat to track stress levels 18
4.2 Why a feedback loop is better than just music 18
4.3 Using chrome to detect the heart beat 18
4.4 Mapping heart data to nature sounds 19
4.5 Regaining control over the internal state 20

5 Conclusion **21**

Quellen **22**

1 Introduction

1.1 The problem: The rising need for non-invasive stress regulation

In our modern society, the prevalence of stress has increased substantially. Recent surveys indicate a significant upward trend: In 2009 31% of workers stated that they are experienced high levels of stress. This number rose to 40% in 2024 (Gallup 2025, 18). Similarly, the german health insurance company Techniker Krankenkassen (TK) indicates that 66% of people report daily stress, compared to 57% in 2013 (Techniker Krankenkasse 2025, 51). These numbers differ from survey to survey, but they display a clear trend: Stress Levels in our society are rising.

There are different forms of stress. Acute stress is experienced for a short time when you have to make an urgent decision. For instance, reacting to a serious traffic situation. In contrast, chronic stress is experienced for a longer time period. Often for weeks or months. Stress has a wide variety of causes. It arises when an individual perceives a situation - such as financial instability or family issues - as a persistent threat. Symptoms include headachs, stomach issues, chest pain, indigestion and a variety of other physical manifestations (Harvard Health Publishing 2024).

According to MCEwen (2017), the brain is the central organ that perceives stress and handles accordingly in the body. It has a direct impact on the cardiovascular system and thereforce influences the heart rate. Consequently, it is important to monitor these systems to detect whether a person is experiencing high levels of stress(McEwen 2017, 2–3).

There are many different ways to measure these cardiovascular signals. Traditional methods include contact-based sensors, such as chest straps or pulse oximeters. These are very precise but on the other hand impractical during daily use. To solve this issue, remote Photoplethysmography (rPPG) proposes a solutions to these limitations. This technology uses standard video cameras to monitor the skin color and filter for subtle skil-color variations that are caused by flowing blood. With the information the heart rate can be calculated (Verkruysse et al. 2008).

1.2 The concept of Biofeedback

Biofeedback describes a therapeutic concept in which a user receives real-time information about their physiological processes, such as respiration or heart rate. This data is typically made accessible through an output device, such as a visual display. Alternatively, the information can be translated into different forms, such as auditory signals. The primary goal of biofeedback is to make these biological

processes perceptible, enabling the user to gain conscious awareness and control over functions that were previously automatic (Giggins et al. 2013).

1.3 The concept: Music lowering stress levels

Music is a powerful emotional stimulus that directly influences the human nervous system. Beyond its aesthetic value, it can create deep feelings and modulate physiological states. A meta-analysis by Pelletier(2004) came to the conclusion that music decreases stress-induced arousal, though the effect depends on indivdual factors like musical preference and experience (Pelletier 2004, 1).

1.4 Objective of this thesis

This thesis evaluates the combination of rPPG and music. First, the user's heart rate is measured using rPPG. This data is subsequently reported back to the user with a biofeedback loop. As discussed previously, music can lead to a significant decrease of stress levels. Therefore, the heart rate is transformed into an auditory signal that the user can listen to. The central objective of this thesis is to analyze the feasibility of such a system and to assess its effectiveness in reducing stress.

1.5 Structure of this Thesis

In Chapter 2 the theoretical foundations are described. It explores the physiological basis of stress and the mechanics of rppG and continues with the concepts of interception and the regeneration that occurs when the heart enters a state of resonance. Chapter 3 reviews the state of the art in musical biofeedback systems, followed by an argumentation in Chapter 4 that synthesizes these findings to evaluate the potential of camera-based systems for consciousness and stress regulation.

2 Theoretical Foundations

2.1 Stress and the Autonomic Nervous System (ANS)

The Autonomic Nervous System (ANS) consists of the parasympathetic and the sympathetic branches. The paraympathikus system act during periods of rest and calm and regulates functions such as "rest and digest". In contrast, the sympathetic system is activated situations in which we precive a threat. In these cases, the heart rate increases to release more energy (McCorry 2007).

McEwen (2007) describes this as the allostatic effect. The body dynamically adjusts its internal state to meet the standards of the environment. For instance, in a life threatening situation a lot of energy is released. The short-term mobilization of energy is a protective reaction of the body. If this activation occurs chronically without enough recovery, it can lead to permenant damage(McEwen 2017).

Individual differecnes in stress perception are rooted in the brains' ability ot evaluate environmental stimuli. According to McEwen (2007), the brain exhibits significan platicity, meaning its responses to stressors is shaped by genetics, early development, and past experiences. If an individual has for example experiences a trauma, the brain's system may become hypersensitiv. It might trigger a 'fight or flight' response in harmless situations. This can lead to coping strategies, such as smoking or drinking. These coping stragies try to calm down the parasympathetic branch of the autonomic Nervous System. On the other hand, the plasticity also allows for an intervention in the other direction. The brain can be trained to shift from a threatening state to a calm state (see Figure 2.1) (McEwen 2017).

Stress has different forms. Good stress describes the process of overcoming a small obstacle and to have a positive outcome. Tolerable stress means that a person with a health brain situation can cope with a critical situation. Often the support of peers or family and friends is necessary. Toxic Stress on the other hand, describes a situation where a person faces a huge obstacle and has no way to overcome this hurdle. These situations tend to results in a higher degree of allostatic overload (McEwen 2017).

2.2 Heart Beat and Heart Rate Variability

To understand the mechanics of stress detection, it is essential to distinguish between the quantity and the quality of cardiac activity. These two aspects are represented by Heart Rate (HR) and Heart Rate Variability (HRV)(Shaffer and Ginsberg 2017).

Heart Rate (HR) is defined as the total number of heartbeats per minute. It serves as a measure of how fast the heart is beating at any given moment. While HR provides

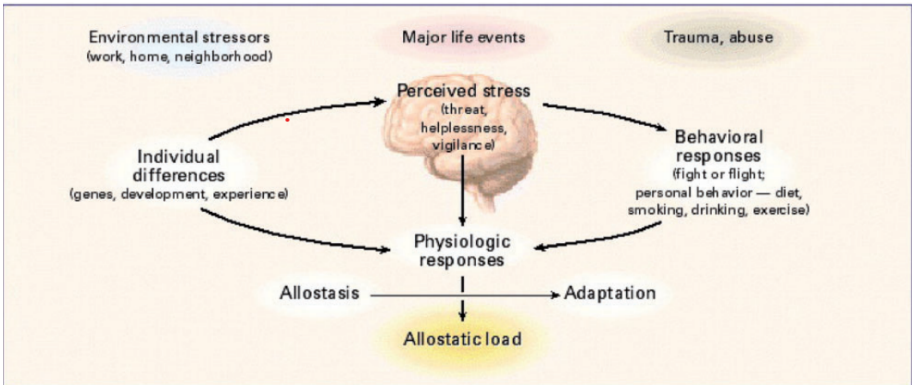

Figure 2.1: The picture describes how stress is created and perceived Source: (McEwen 2017)

a general overview of physical or emotional exertion, it remains a relatively coarse metric. For instance, in an office environment, HR often stays within a narrow range, which can make it an insensitive marker for subtle cognitive loads or early-stage stress(Shaffer and Ginsberg 2017).

Heart Rate Variability (HRV), on the other hand, focuses on the micro-level of cardiac timing. It refers to the fluctuation in the time intervals between consecutive heartbeats, known as Interbeat Intervals (IBIs). HRV consists of the constant changes in these time intervals from one beat to the next(Shaffer and Ginsberg 2017).

From a physiological standpoint, a healthy biological system requires a high degree of adaptability. This means that a healthy heart does not maintain a perfectly rigid or identical time between intervals. Instead, a resilient autonomic nervous system constantly fine-tunes the timing of each beat to respond to internal and external demands(Shaffer and Ginsberg 2017).

If the intervals between heartbeats become too consistent and lose this natural irregularity, it is often a clinical sign of physiological strain or a lack of regulatory flexibility. High variability indicates that the body is in a state of balance and can rapidly adjust to stressors. Therefore, while HR measures the quantity of beats, HRV reflects the quality and efficiency of the autonomic nervous system's control over the cardiovascular system (Shaffer and Ginsberg 2017).

2.3 Remote Photoplethysmography (rPPG)

2.3.1 The foundations of rPPG

The foundational work for modern remote photoplethysmography (rPPG) was made by Verkruysse et al. (2008), who demonstrated that the heart beat could be monitored using a standard digital camera. This relies on the fact that hemoglobin absords light differently than the surrounding body tissue. As the blood volume changes in the body with every heartbeat, the intensity of the reflected light varies. Their research identified that the green color channel contains the strongest information regarding the heart beat (see Figure 2.2). Despite the successful proof-of-concept,

two challenges were highlighted: Motion artifcats caused by movement and noise due to the camera's sensor (Verkruysse et al. 2008).

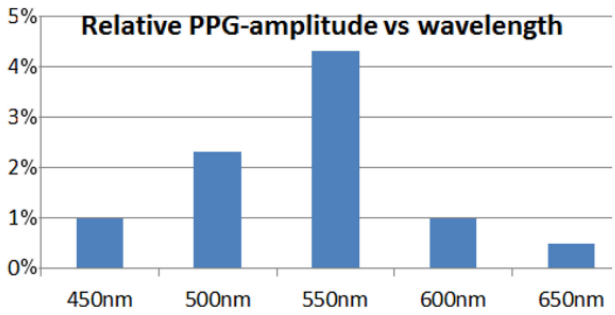

Figure 2.2: Spectral sensitivity of the PPG signal. Green light exhibits the highest amplitude, making it the most reliable channel for non-contact heart rate monitoring Source: (Haan and Jeanne 2013)

2.3.2 Evolution: From Classic Algorithms to AI-based Detection

Following the initial proof-of-concept, a significant increase in rPPG research was observed, particularly after 2017. This trend is closely linked to the rapid advancements in Artificial Intelligence (AI) and Deep Learning. A recent meta-analysis revealed that in 2025 82.05% (Debnath and Kim 2025) of papers employed an approach with artificial intelligence. A major problem is the availability of diverse, large-scale datasets. Current sets only have data of people from North America, Europe and China. This leads to a severe underrepresentation of various age groups and diverse skin tones (Debnath and Kim 2025).

2.3.3 Tradtional Frameworks and Region of Interest (ROI)

Traditional rPPG frameworks typically follow a multi-stage process. First, the face is detected and tracked, often using the Viola-Jones algorithm or similar cascade classifiers to define a Region of Interest (ROI). Then methods such as Principal component Analysis (PCA) or Independet Component Analysis (ICA) or employed to isolate the pulse signal from the green color channel. While effective if the person remains still, these approaches struggle with motion artifacts, as they require several seconds of stable data to compute the signal components (Debnath and Kim 2025).

While traditional frameworks rely on fixed mathematical models, modern approaches increasingly replace stages of the rPPG pipeline with Deep Learning modules. This is done in the area of face detection and tracking. The Viola-Jones algorithm often struggles with varying lighting conditions and extreme head poses. Current systems frequently utilize more robust AI-based detectors (Debnath and Kim 2025).

Furthermore, the Region of Interest (ROI) significantly impact the accuracy of the heart rate. Although, early studies often used the entire facial area, recent studies recommend using a small rectangular area at the forhead for better results (Debnath and Kim 2025).

2.3.4 Chrom approach

The Chrominance-based method (CHROM) introduced a physicsbased alternative. It solves the major challenge of subject motion, when calculating the heart beat. Periodic movements can render simple intensity-based selections useless. Furthermore, while many algorithms require long observation intervals to achieve sufficient frequency resolution, CHROM is designed to be highly efficient. Instead of minimizing brightness variations through simple division, CHROM uses a subtractive approach. By subtracting specific wavelength components, common-mode noise is mathematically cancelled out, as shown in the simplified logic:

$$(R + noise) - (G + noise) = R - G$$

The CHROM algorithm projects the normalized RGB signals into two orthogonal chrominance signals, X and Y. In a perfect environment under white light, these signals are mathematically independent. Based on empirical data from a diverse set of subjects, de Haan and Jeanne derived the standardized projection vectors $[0.7682, 0.5121, 0.3841]$, leading to the core transformations:

$$X_s = 3R_n - 2G_n$$

$$Y_s = 1.5R_n + G_n - 1.5B_n$$

To further refine the pulse signal S, these components are combined using the ratio of their standard deviations (α):

$$S = X_s - \alpha \cdot Y_s \quad \text{where} \quad \alpha = \frac{\sigma(X_s)}{\sigma(Y_s)}$$

Even after projection, the signal S may contain a slow "drift" caused by respiration or minor posture changes. To isolate the cardiac signal, a Bandpass Filter is applied. The filter is typically tuned to a range of 0.7 Hz to 4.0 Hz, which corresponds to a human heart rate between 42 and 240 BPM. Frequencies outside this window are discarded as non-physiological noise.To extract the final heart rate, a Fast Fourier Transform (FFT) is performed on the filtered signal. The FFT shifts the data from the time domain to the frequency domain, resulting in a power spectrum. The algorithm identifies the "dominant peak" (the highest frequency component) in this spectrum. For instance, a peak at 1.2 Hz is converted to the final pulse rate:

$$1.2 \text{ Hz} \times 60 \text{ seconds} = 72 \text{ BPM}$$

One of the most significant advantages of CHROM is its temporal efficiency. While traditional BSS methods (Blind Source Separation) like ICA or PCA often require up to 30 seconds of data to stabilize, CHROM utilizes an Overlap-Add (WOLA) procedure. By using overlapping windows of only 1.6 seconds, the system provides a continuous, near-instantaneous pulse estimate.

In comparative studies, the robustness of CHROM is evident:

- Stationary Accuracy: CHROM achieves a 98% success rate in predicting the correct pulse, with a 92% correlation to medical-grade contact sensors. The Root Mean Square Error is impressively low at 0.4 BPM.
- Motion Resilience: Under heavy motion (e.g., on a fitness stepper), CHROM maintains a 48% success rate, whereas ICA and PCA fail significantly, dropping to 4% and 11% respectively. This is because BSS methods often "guess" which signal component represents the pulse, frequently choosing the motion artifact instead.
- Skin Tone Diversity: While darker skin types (high melanin content) naturally lower the Signal-to-Noise Ratio due to higher light absorption, CHROM remains the most robust classical algorithm across all phenotypes, maintaining functionality where other methods cease to operate (Haan and Jeanne 2013).

2.3.5 Modern Deep Learning Paradigms

With the rapid advancement in artificial intelligence, modern rPPG systems have shifted towards end-to-end models. These architectures can process raw viode data or image sequences to estimate heart rate directly, often wihtout the need for manual face isolation or Region of Interess (ROI) selection. There are three architectural paradigms: Convolutional Neural Networks (CNNs), Hybrid Neural Networks and Transformed-based architectures (Debnath and Kim 2025).

In modern rPPG systems, Convolutional Neural Networks (CNNs) are the standard for processing image data. These networks consist of several layers, mainly an input, an output, and various hidden layers like convolutional and pooling layers. Their main task is to recognize the very subtle color changes in the skin pixels that happen during a cardiac cycle.

A big problem in this field, is that there aren't enough large and diverse datasets to train these models from scratch. Because of this, researchers often use Transfer Learning (Debnath and Kim 2025). The idea here is to take a model that was already trained on a different task, such as general face recognition. By 'freezing' the first layers and only fine-tuning the last ones with rPPG-specific data, the model can use its existing knowledge of facial features. This approach makes it possible to get good results even with smaller datasets and saves a lot of training time (Pan and Yang 2009).

In addition, a different category of neural networks has emerged: Hybrid Neural Networks. These combine different neural networks like CNNs, RNNs and autoencoders to form a more robust system (Debnath and Kim 2025).

In the past years powerful Large Language Models (LLMS) like ChatGPT have emerged that use transformers to understand words and sentences. Previously, it was a huge problem for models to pick up the order of words and the context. These transformers were able to overcome this hurdle and understand text better by breaking it up into their individual words and syllables. The same technique can be employed to estimate the heart beat. The face is broken up into small rectangular

patches and, which the network analyzes in relation to one another. This helps the model to understand the global context of the face (Dosovitskiy et al. 2021). By Comparing different regions simultaneously, the transformer is able to differentiate between the heart beat and noise, such as micro-movements. This leads to a more robust heart rate estimation (Debnath and Kim 2025).

2.3.6 Summary of rPPG Frameworks

There are lot of different approaches to calculate the heartbeat with an rPPG technique. Modern Deep Learning Algorithms have emerged in the recent years. They show promising results, although there are not enough diverse datasets. The CHROM Algorithms is promising for a realtime evaluation of data. Traditionally algorithms like PCA are slow and not so precise. Although, there has been a huge advancement in the past years, there are still no long-term studies that monitor the heart beat over several hours. Most researches only test rPPG for a short time, which does not show how it works in real daily life (Debnath and Kim 2025).

2.4 Music and Stress Reduction

While Deep Learning and tradtional approaches like CHROM provide the functionality to capture the heart beat remotely, the effect still depends on the conversion of this data to music. This leads to the question how exactly the heart beat should be converted to a piece of music.

There are no clear results regarding music and its effects on stress reduction. Results differ from study to study. According to a meta-analysis by Pelletier(2004), music consistently assists in lowering stress levels. The magnitude of this response is affected by several factors. Patients with prior musical experience, such as playing an instrument, tend to lower their stress levels a lot more. Furthermore, Pelletier notes that indivudal therapy generally results in better relaxation than group session. Although, group sessions are used much more often because they are less expensive. A huge obstacle is that the devices that are used for data collection often cause discomfort. This can lead to false data regarding the relaxation process. Finally, the study suggests that music based on scientific research (e.g., specific rhythmic or frequency patterns) is more effective for stress reduction that music chosen soley based on personal preference(Pelletier 2004).

According to Koelsch and Jäncke (2015) the emotions and the heart activity are closely linked. For instance, at a frisson the heart rate is rising. They also came to the conclusion that the effects of music on the heart are small and that results of studies on this topic are often inconsistent. They point out that, there is stil lack of high quality research on this topic (Koelsch and Jäncke 2015).

In another study, Linnemann et al. examined how music affects stress in daily life rather than in a laboratory. They came to the conclusion that listening to music can effectively reduce stress levels. Nevertheless, the largest effect is reached when subjects listen to music with then intention of decreasing stress levels. This suggests

that the reason why someons listens to music is very important for the actual stress-reducing effect (Linnemann et al. 2015). These results are also supported by another study (Thoma et al. 2011). They confirm that listening to music alone does not automatically decrease stress levels. Instead, the reason of listening to the music is the most important factor. For example, if someone listens to music to cope with loneliness, it might effectively decrease their feeling of isolation. Nevertheless, if the goal is relaxation, the music only works well if the listener has the specific intention to relax. The study of Bernardi et al. (2006) confirms that the results do not depend on the music preferation of the user rather on the BPM and rhythm structure of the music. Additionally, the study found no evidence of habituation, meaning the users do not simply 'get used' to the music over time. In Contrast to Linnemann et al. (2015), who emphasized the importance of the listener's intention, Bernadi concludes that music can be perceived as a passive form of meditaiton. consequently, this means that the user does not need an intend to reduce stress for measurable results (Bernardi et al. 2006). Research by Nater et al. (2006) unterlines the importance of considering sex differences when using music for emtion induction or stress management. They suggest that women react more intensely to music they do not linke. In general, the findings indicate that women are more stress-reactive than men in response to certain musical triggers. Music is also a tool for invoking emotion changes. They also underline that music that the user prefers does not automatically lead to relaxation (Nater et al. 2006).

2.5 Consciousness and State Regulation

Consciousness, in the context of self-regulation, is frequently described through the concept of mindfulness. This is the process of directing one's attention to the present moment. Often one's focus is directed on signals such as the respiration or heart rate. By using this strategy, individuals develop a sensitivity to their immediate experience, including thoughts and emotional states. A fundamental requirement of this operational definition, as established by Bishop et al. (2004), is a non "non-judgmental" approach. This means that internal states are perceived with curiosity and openness. Internal thoughts and feelings should be accepted as they are (Bishop et al. 2004).

Research has demonstrated that the practice of mindfulness significanlty enhances emotion regulation and reduces perceived stress. According to Tang et al. (2015), these benefits apply to both physical and mental health and also improve cognitive regulation. Consequently, mindfulness has to potential for the treatment of various clicnical disorders. However, it must be noted that despite these promising findings, the current state of research remains limited. There are still not enough studies to provide long-term evidence regarding the state of mindfulness (Tang et al. 2015).

To fully grasp the necessity of minduflness, it is essential to understand the state of the human body when mindfulness is not applied. Robert J. Aumann describes this state as a form of 'autopilot', where complex behaviors - such as driving a car- become a habit. This effective is useful for routine task. On the other hand, it becomes detrimental when applied to stress. In its acute, stress is a negative emotion, intended to activate an adaptive response. Consequently, under chronic conditions,

this signal often becomes a habit and is always running in the nonconscious mind. As a result, the human body no longer receives this a conscious warning, leading to permanant damage(Aumann 2024) .

This transition from an acute warning to a chronic habit marks a breakdown in the body's internal communication system. To understand how the 'autopilot' manages to silence these vital signals, one must look at the underlying mechanism of interoception.

Interoception is the fundamental process of receiving, accessing, and appraising internal bodily signals. It acts as a primary motivator to maintain our internal balance and directly shapes our moods and decision-making processes. In the brain, specifically the insular cortex, all these signals—heartbeat, pressure, temperature—converge. However, in a high-stress environment like a modern office, these signals become noisy or blurred. The results of the bodily processes still reach the brain, but the the insula is overwhelmed and can no longer translate them into meaningful information.

To understand why a biofeedback system is necessary for everyone, we must look at the four pillars of interoceptive integration:

- Interoceptive Accuracy: The objective, measurable ability to precisely detect a signal (e.g., counting your exact heart rate).
- Interoceptive Sensibility: The subjective belief or "feeling" that one is good at sensing the body.
- Coherence: The alignment between a biological state (like low blood sugar) and the resulting feeling (like irritability).
- Regulation: The skill to actively "catch" and modulate a signal, such as slowing down a racing heart through breathing.

Crucially, research shows that even people who meditate are not necessarily "better" at objective Accuracy. They might have a higher attention tendency, but they struggle just as much with precise detection. This is why a technical biofeedback system necessary as it provides the objective data that even experts miss.

The real danger in stress is the split in our neural networks. We often get stuck in the conceptual Network, which is busy analyzing, judging, and ruminating about the past or future. This drowns out the sensory Network, which is responsible for the immediate, felt experience of the current moment. When this happens, interoceptive signals become dangerous. We might catastrophize a simple fast pulse as a heart attack because we lack the sensory clarity to interpret it correctly (Farb et al. 2015).

2.6 Biofeedback-Driven Regeneration

To understand how a Biofeedback-driven-system can lead to regeneration in the human body, it is essential to examine the underlying biological closed-loop systems.

According to Lehrer and Gevirtz (2014), the efficacy of heart rate variability Biofeedback relies on the synchronization of the cardiovascular and respiratory systems through several key pathways(Lehrer and Gevirtz 2014).

2.6.1 The Principle of Resonance Frequency

The human cardiovascular system functions as an oscillating physical system with a specific resonance frequency. For most adults, this frequency occurs at approximately 0.1 Hz, corresponding to a breathing rate of roughly six breaths per minute(Lehrer and Gevirtz 2014).

In a resting or stressed state, heart rate patterns are often chaotic due to competing autonomic influences. Upon reaching resonance, these patterns transform into a clean, high-amplitude sinusoidal curve. This represents a state of maximum internal order and physiological efficiency. This frequency is influenced by total blood volume. Therefore, taller individuals and biological males often exhibit a slightly lower resonance frequency due to larger vascular capacities(Lehrer and Gevirtz 2014).

2.6.2 Autonomic Homeostasis and the Baroreflex

The primary regulatory mechanism for maintaining biological equilibrium is the baroreflex. It controls the internal blood pressure. By breathing at the resonance frequency, heart rate and blood pressure oscillations enter a perfect phase relationship. Rather than working against each other, they synchronize to maximize the baroreflex gain. In this synchronous state, gas exchange in the lungs becomes highly efficient. The organism receives maximum oxygenation with minimal metabolic effort. This efficiency is the fundamental biological basis for physical regeneration(Lehrer and Gevirtz 2014).

2.6.3 The Vagus Nerve and the "Braking System"

The vagus nerve is communicate between the hart and the blood. A physical pathway to transmit these regulatory signals is required. This role is fulfilled by the vagus nerve, the longest nerve of the nervous system and the primary component of the parasympathetic system. During resonance breathing, the activity of the vagus nerve is rhythmically stimulated. This strengthening is what allows the body to lower the heart rate more effectively(Lehrer and Gevirtz 2014).

2.6.4 Cognitive Anchoring and Altered Consciousness

Beyond biochemical changes, Biofeedback induces a measurable meditative effect that alters the user's state of consciousness. The requirement to focus on the biofeedback signal acts as a cognitive anchor. Because working memory has a limited capacity, the brain cannot simultaneously process anxious worrying while maintaining the precise respiratory control needed for resonance. (Lehrer and Gevirtz 2014).

3 State of Art: Musical Biofeedback Systems and Empiral Evidence

3.1 Case Study: The "Unwind" Musical Biofeedback System

3.1.1 Concept and Design Philosophy

The "Unwind" system, developed by Yu et al. (2018), represents a sophisticated approach to musical biofeedback designed specifically for relaxation assistance. In the development of the interface, the researchers found two primary methods of presenting cardiac data through audio. The first involves direct modulation of a song's Beats Per Minute (BPM) to reflect the heart rate. However, experimental data indicated that this method can be counterproductive. Participants frequently reported an increased psychological burden and feelings of anxiety due to the abrupt and mechanical changes in musical rhythm. In other cases, the predictability of the BPM-mapping led to boredom and mental fatigue.

To avoid these issues, Unwind utilizes a layered approach where biofeedback is integrated into a nature soundscape combined with sedative music. This ensures that the output is not only an informative data display but also remains a well-composed, aesthetically pleasing piece of music. Sedative music is a cornerstone of music therapy, known to effectively decrease heart rate and blood pressure through its low tempo and repetitive structure.

3.1.2 The Multi-Layered Mapping Strategy

The core of "Unwind" lies in its dual-layer mapping strategy, which separates immediate physiological changes from long-term relaxation trends.

3.1.2.1 The Information Layer: Respiratory Feedback

The system uses the Inter-Beat-Interval (IBI)—the time between consecutive heartbeats—to provide real-time feedback on the user's breathing. The heart rate accelerates during inhalation (decreasing the IBI) and decelerates during exhalation (increasing the IBI). Unwind maps these fluctuations directly to a wind soundscape. During Inhalation the wind volume is reduced and during exhalation the wind volume becomes louder. This creates an intuitive mapping where the user literally hears their own breath through the wind. The goal is for the user to regulate their breathing until the wind sound is like a smooth, consistent sine wave, promoting consistent respiration.

3.1.2.2 The Evaluation Layer: HRV and Nature Sounds

While the wind provides immediate feedback, the overall relaxation success is mapped to the complexity of the nature soundscape using a specialized form of Heart Rate Variability. The system calculates variability using a sliding window of 16 heartbeats.

The window size of 16 beats is a deliberate technical choice. It is large enough to capture at least one full respiratory cycle, which prevents hectic or fragmented audio changes. At the same time, this interval remains small enough to stay sensitive to subtle shifts in the user's physiological state.

The organic soundscape is modulated based on the intensity of the Heart Rate Variability (HRV). If the HRV is low the acoustic environment becomes busy and crowded with a high density of bird calls and geophysical noises. Conversely, if the variability increases, the soundscape becomes simpler and more transparent. The frequency of animal sounds decreases and the overall volume drops, rewarding the person with acoustic clarity. Through this transition toward silence, the user is guided into a deeper state of relaxation.

3.1.3 Experimental Results and Validation

The system was validated with 40 young adults (22 females, 18 males) using a rigorous procedure involving a baseline phase, a stress-inducing phase (Stroop color-word test), and a subsequent relaxation session. The study utilized both physiological measurements (Heart Rate, Respiration Rate, Skin Conductance) and self-reports. The findings were significant. A significant decrease in heart rate was found in the music-assisted conditions, and the reduction in respiration rate was significantly greater with biofeedback than without. Skin conductance responses, which indicate sympathetic nervous system activity, decreased markedly during the relaxation sessions. In follow-up interviews, 70% of participants reported that the nature sounds allowed them to quickly visualize calming natural scenes, aiding their focus and mood.

Ultimately, the combination of sedative music and nature-based biofeedback was perceived as the most effective and "smoothing" experience. By providing an "information layer" that does not disrupt the "musical layer," Unwind proves that biofeedback can be both medically informative and psychologically restorative (Yu et al. 2018).

3.2 Analysis of Music as a Biofeedback Signal

3.2.1 Study Design and Methodology

In their pivotal study, Bergstrom et al. (2014) investigated the efficacy of music as a carrier for biofeedback signals. The researchers compared three distinct experimental conditions to determine their impact on physiological arousal. The study compared

three conditions: (A) Music Only (pre-recorded tracks), (B) Sonification (sine-wave pitch reflecting HR), and (C) Combined Music Biofeedback (musical tempo and volume modulated by HR).

To ensure scientific neutrality, the study utilized algorithmically composed music. This approach avoided the sound of popular songs, using neutral classical instrumentation without lyrics. A sedative version with a base tempo if 60 BPM for relaxation and an activating version with a base tempo of 140 BPM for arousal.

3.2.2 Technical Implementation: Mapping and Smoothing

A critical challenge in musical biofeedback is maintaining musicality while conveying data. To prevent hectic jumps in the audio, the system employed a moving window average of the last four heartbeats. The heart rate measured via a chest strap was then linearly mapped to the tempo and volume of the music. This ensured that the feedback remained a constistent piece of music rather than just a series of data-driven noises.

3.2.3 Key Findings and Participant Behavior

The study required participants to actively perform two tasks. One task was used to maximize arousal (Task UP) and another to minimize arousal (Task Down). While participants in all conditions were successful in altering their heart rate, the music biofeedback (Condition C) was perceived as significantly more effective and engaging than music alone.

Interestingly, although no instructions were given on how to change their heart rate, participants intuitively used respiration as a primary strategy. By breathing faster or slower, they managed to change the musical signal. This identifies musical biofeedback as a highly intuitive tool for self-regulation.

3.2.4 Separating Respiration from Heart Rate

The most significant scientific contribution of the study lies in the path analysis. Since breathing naturally influences heart rate, the researchers statistically removed the effect of respiration to see if the biofeedback had a direct physiological impact.

The results revelaed that in Condition A (Music Only) the effect of music alone on the heart rate was not significantly different from zero. This suggests that passive music listening relies almost entirely on the “detour” through the lungs. On the other hand, in Condition B & C (Biofeedback) the results showed a highly significant direct effect on the heart rate.

3.2.5 Conclusion: Why Music Matters

The study concludes that biofeedback is necessary for a person to achieve a direct physiological response to their intention to relax or arouse. While pure sonification (Condition B) is effective, musical biofeedback (Condition C) offers distinct advantages for long-term use, such as in an office environment. Music is more effective at drawing and maintaining the user's focus. The aesthetic variation in music prevents the "fatigue" and annoyance caused by repetitive sine-wave tones (sonification). Unlike pure sonification, music prioritizes the user's emotional experience, making the biofeedback a pleasant rather than a functional task (Bergstrom et al. 2014).

3.3 Case Study: Technology-Mediated Mindfulness

3.3.1 Problem Statement and Objectives

In the modern workplace, traditional mindfulness practices often fail not due to a lack of interest, but due to practical barriers. Balconi et al. (2017) identify the high demand for consistent discipline and the absence of immediate motivation as the two primary obstacles for mindfulness. Since emotional stress responses are often rooted in automatic mechanisms, beginners struggle to access their internal states without external assistance. The objective of this study was to investigate whether a technology-mediated intervention could lower the entry barriers of classical meditation and produce measurable improvements in both emotional regulation and cognitive performance.

3.3.2 Methodology and Experimental Design

The researchers conducted a four-week intervention study involving 40 participants suffering from mild-to-moderate stress. The participants were divided into an experimental group and an active control group. This experimental group utilized the Muse™ headband, a mobile EEG system connected to a smartphone application. The participants were tasked with a daily, active attention-training session. The device acted as an objective real-time feedback instrument. Brainwaves were translated into acoustic signals (weather sounds). A calm mind produced gentle, peaceful sounds, while wandering or stressed thoughts triggered storm and rain sounds. The active control group performed identical relaxation exercises for the same duration but without the wearable device or acoustic feedback. They acted as passive observers without objective data regarding their mental state.

This design is methodologically crucial because it isolates the variable of technological mirroring. It answers the question whether the success due to the time spent relaxing, or due to the real-time feedback provided by the technology.

3.3.3 Muse headband

The Muse™ headband is a mobile neurofeedback system that monitors cortical electrical activity (EEG) via seven dry sensors. Its core functionality is the real-time sonification of brainwaves: a smartphone application translates the user's mental state into an adaptive nature-based soundscape. While calm, focused states are rewarded with peaceful ambient sounds and birds chirping, increased stress or mind-wandering triggers turbulent weather sounds like wind and rain. This acoustic feedback serves as a real-time indicator, enabling users to perceive unconscious mental states and actively regulate them through breathing and focus(see Figure 3.1).

Figure 3.1: Subject wearing the Muse™ EEG headband during a neurofeedback session. Source: (Balconi et al. 2017)

3.3.4 Analysis of Results: Psychological and Cognitive Effects

The results after four weeks were significant and underscore the potential of wearables in professional stress management. While the control group showed negligible changes, the experimental group recorded a decrease in perceived stress (measured via the PSS-10 scale) of 7% after two weeks and a total of 16% by the end of the four-week period. Beyond mere relaxation, executive functions improved. Participants demonstrated higher concentration levels and better impulse control during complex tasks. This was electrophysiologically evidenced by an enhancement of the N2 amplitude in the EEG—a known marker for focused attention orientation. The participants' brainwaves showed a lasting shift toward a relaxed mindset. This suggests that the system triggers actual neural plasticity, rather than just providing a temporary "quick fix."

3.3.5 Conclusion: Technology as a Motivational Catalyst

The central finding of Balconi et al. is that real-time feedback is the decisive factor for success. The technology serves as a motivator that makes otherwise invisible progress tangible through statistics, rewards, and milestones. This addresses the boredom barrier of traditional meditation, as the user develops a sense of agency.

For the rPPG-based system proposed in this thesis, Balconi's study provides the necessary empirical foundation. It proves that transforming physiological data into acoustic stimuli does not just promote relaxation—it actively trains the brain to maintain high cognitive performance under pressure (Balconi et al. 2017).

4 Discussion

4.1 Using the heartbeat to track stress levels

The physiological foundation of this system rests on the fact that stress is not just a feeling but a measurable state of the Autonomic Nervous System. While Heart Rate provides a basic metric, it is the Heart Rate Variability (HRV) that really reflects the battle between the sympathetic and parasympathetic branches. In a high-stress office environment, the bodys adaptability decreases. This leads to a rigid, metronomic heartbeat with low HRV. Because these changes occur on a micro-level of milliseconds, they serve as the primary data input for a stress-regulation system, long before the user may conciously feel stressed or tired.

4.2 Why a feedback loop is better than just music

The main challenge in chronic stress is the "autopilot" effect described by Aumann (2024). When stress becomes a habit, the brain's internal monitoring tool loses its ability to distinguish signals from noise. This breakdown in interoception means that even if a person intends to relax, they lack the objective reflection to see if their efforts are working.

It is important to note that passive music listening alone often fails to break this cycle. While relaxing music can lower arousal, the effect is often inconsistent because the user remains a passive listener. Without a feedback loop, the "autopilot" stays in control and the person has no way to see if the music actually changes their physiology. A biofeedback system is therefore necessary to externalize these internal signals. A biofeedback system is therefore necessary to externalize these internal signals, creating a closed-loop interaction between physiology and sound (see Figure 4.1).

4.3 Using chrome to detect the heart beat

To close this loop without the discomfort of physical sensors—which can themselves induce stress—the rPPG technique is used. Among various frameworks, the CHROM algorithm is uniquely suited for this task. Its mathematical simplicity allows for a low-latency loop, which is crucial because any delay between a heartbeat and its musical representation would break the users sense of Agency.

While modern Deep Learning models like CNNs show high accuracy in labs, they often need a lot of GPU power which would slow down a normal office computer or interfere with other tasks. Also, many models have a bias because they are trained on

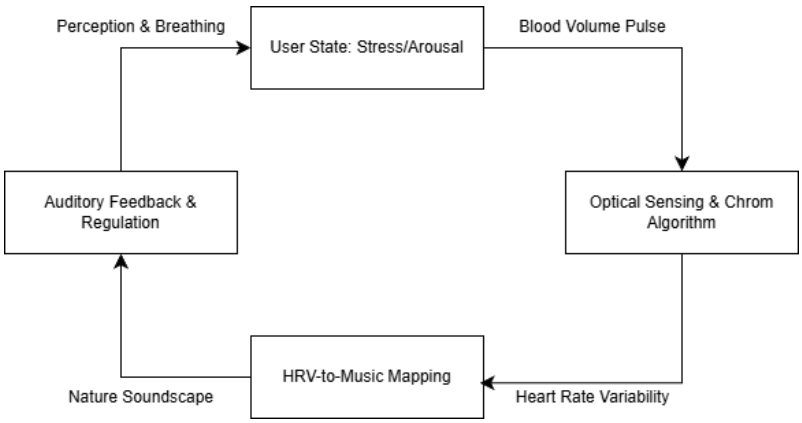

Figure 4.1: Schematic representation of the closed-loop musical biofeedback system using rPPG and the CHROM algorithm

limited datasets. This means they often fail on different skin tones, while CHROM is a physic-based approach and remains more inclusive. Another big factor for the biofeedback is the real-time capability. Many AI models need to look ahead at future frames to stabilize the signal, but for a live feedback loop, we need the instant response that only a simple, fast algorithm like CHROM can provide. By filtering out motion artifacts and remaining robust across different phenotypes, CHROM ensures that the data acquisition stage of the loop remains stable even if the user moves while typing or working.

4.4 Mapping heart data to nature sounds

The translation of the heartbeat into a nature-based soundscape is the bridge between machine data and human emotion. Unlike pure sonification with simple beeps, which causes fatiuge and boredom, an adaptive soundscape of birds, wind, or rain engages the users Sensory Network. By mapping a low HRV to a crowded or busy acoustic environment and rewarding a high HRV with silence and clarity, the system creates an intuitive incentive.

Research shows that nature sounds help the user to quickly visualize a calm scene in their mind, which acts as a secondary relaxation trigger. This is more effective than just changing the BPM of a song, because if the music mapping is too mechanical, the person gets annoyed or tired. According to Bernardi et al., the cardiovascular system naturally synchronizes with musical rhythm, so the nature sounds serve as an information layer that guides this process. The user does not just see a graph. They feel the congestion of their stress and the relief of their relaxation. By externalizing the heartbeat into these sounds, the system provides additional resources for the brain.

4.5 Regaining control over the internal state

The final stage of the loop is the return to human intentionality. As the user hears the melodic reflection of their own heart, they can begin to reflect with the soundscape. By intentionally slowing their respiration the user directly influences the music. This creates a synergy where the machine provides the data and the human provides the will.

The audible feedback allows the user to calibrate their self-awareness again. This process shifts the brain from the ruminative Conceptual Network back to the restorative Sensory Network. This does not work on its own. The efficacy comes from the users active engagement with the sound. Ultimately, the system functions as a tool for conciousness regulation, helping the person to re-establish their focus and steer their biological state back to a healthy state.

5 Conclusion

The results of this thesis demonstrate that a musical biofeedback system based on camera-based rPPG is not only a theoretical concept but a realistic and highly effective intervention for the modern workspace. By combining remote sensing with adaptive soundscapes, the technical and psychological requirements for effective stress management can be met without the need for uncomfortable wearables.

Among the various frameworks evaluated, the CHROM algorithm proved to be the most viable method for this application. Its mathematical simplicity ensures the low-latency processing necessary for a seamless feedback loop, while its resistance to motion artifacts and skin tone bias makes it superior to current deep learning models for real-time office use. This technical stability provides the objective "mirror" that is needed to break the "autopilot" of chronic stress.

Furthermore, the integration of HRV-data into nature-based musical soundscapes offers a clear pathway for both stress reduction and mindfulness. The system effectively externalizes interoceptive signals, allowing users to display their internal state.

In conclusion, a camera-based musical biofeedback system is a feasible and scientifically grounded tool. It acts as a bridge between human-computer interaction and self-regulation, helping users to reclaim their conciousness from chronic stress.

This thesis was created with the template from Carsten Gips(Gips o.D.).

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